Skywork-OR1-Math-7B and DeepSeek-R1-Distill-Qwen-7B do not have distinct Pass@K gaps for large K values. We suspect the reason is that the distilled LLM may already master major reasoning capabilities that can be learned with RLVR using answer correctness as the reward. So in math domains, their main improvements lie in Pass@1. In contrast, for code domains, applying RLVR to distilled LLMs can still stimulate them to fit for real-world execution feedback, thereby incentivizing extended reasoning boundary.

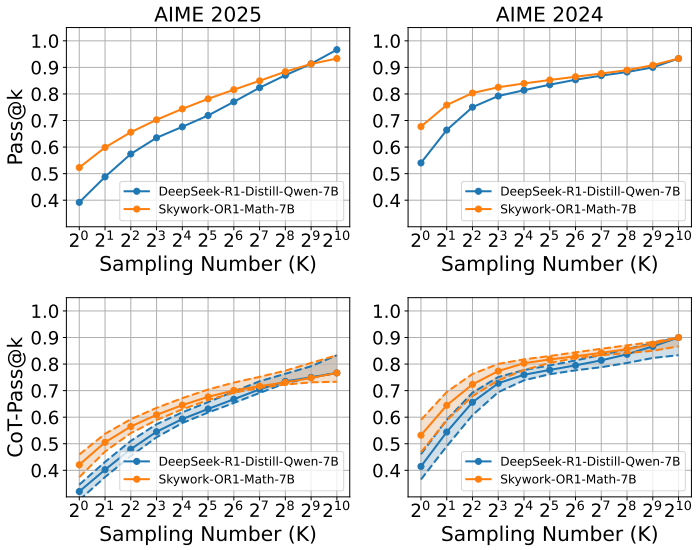

Figure 9: Comparisons of Pass@K (the top row) and CoT-Pass@K (the bottom row) on AIME 2024, 2025 to show how RLVR could improve distilled LLMs. Here the distilled LLM is DeepSeek-R1-Distill-Qwen-7B, and the post-RLVR model is Skywork-OR1-Math-7B. For CoT-Pass@K, we perform multiple verifications for each CoT using DeepSeek-R1-0528-Qwen3-8B, and display the results determined by *any-correct*, *all-correct*, and *majority-correct* strategies, which constitute the shaded area in lower subplots.

A.5 Proof and Additional Theoretical Analysis

Below we include the detailed proof for Theorem 1.

Proof 1 *Let $p_c = P(\mathcal{I}_{CoT}(c_i) = 1)$ be the current probability of generating a correct CoT. The expected reward for a response y_i is:*

$$\mathbb{E}[R(y_i)] = \begin{cases} \alpha & \text{if } \mathcal{I}_{CoT}(c_i) = 1 \\ \beta & \text{if } \mathcal{I}_{CoT}(c_i) = 0 \end{cases} \tag{6}$$

The group-level expected reward $\mu \triangleq \mathbb{E}[\mu_{\mathbf{Y}}]$ is:

$$\mu = p_c \alpha + (1 - p_c)\beta. \tag{7}$$

For large G, the group mean $\mu_{\mathbf{Y}}$ and variance $\sigma^2_{\mathbf{Y}}$ concentrate around their expectations:

$$\mu_{\mathbf{Y}} \xrightarrow{G\to\infty} \mu \tag{8}$$

$$\sigma^2_{\mathbf{Y}} \xrightarrow{G\to\infty} \sigma^2 > 0. \tag{9}$$

The expected advantage conditional on CoT correctness is:

$$\mathbb{E}[\hat{A}(y_i) \mid \mathcal{I}_{CoT}(c_i) = 1] \xrightarrow{G\to\infty} \frac{\alpha - \mu}{\sigma} \tag{10}$$

$$\mathbb{E}[\hat{A}(y_i) \mid \mathcal{I}_{CoT}(c_i) = 0] \xrightarrow{G\to\infty} \frac{\beta - \mu}{\sigma}. \tag{11}$$

Substituting equation 7 into equation 10 and equation 11:

$$\mathbb{E}[\hat{A}(y_i) \mid \textit{correct CoT}] \to \frac{(1-p_c)(\alpha-\beta)}{\sigma} \tag{12}$$

$$\mathbb{E}[\hat{A}(y_i) \mid \textit{incorrect CoT}] \to \frac{-p_c(\alpha-\beta)}{\sigma}. \tag{13}$$

Since $\alpha > \beta$ (by equation 4 under the Logic Prior *assumption) and $\sigma > 0$, we have:*

$$(1-p_c)(\alpha-\beta)/\sigma > 0,$$
$$-p_c(\alpha-\beta)/\sigma < 0,$$

proving inequalities equation 5.

The GRPO policy gradient update in equation 3, $\nabla_\theta J(\theta) \approx \frac{1}{G}\sum_{i=1}^{G} \hat{A}(y_i)\nabla_\theta \log \pi_\theta(y_i \mid q)$, on average increases the likelihood of responses with $\hat{A}(y_i) > 0$ (correct CoTs) and decreases it for $\hat{A}(y_i) < 0$ (incorrect CoTs). Thus, p_c increases monotonically.

Discussions on (μ, σ^2) in Theorem 1 From equation 7, we know that the group reward mean is given by $\mu = p_c\alpha + (1-p_c)\beta$. Furthermore, we can derive the exact formula for the variance σ^2 in equation 9 and analyze their impacts together with p_c, α, and β on policy iterations.

The sample variance $\sigma_{\mathbf{Y}}^2$ converges to the true variance σ^2:

$$\sigma_{\mathbf{Y}}^2 = \frac{1}{G}\sum_{j=1}^{G}(R(y_j) - \mu_{\mathbf{Y}})^2 \xrightarrow{G\to\infty} \mathrm{Var}(R(y_j)) \equiv \sigma^2,$$

where $\mathrm{Var}(R(y_j))$ can be computed using the law of total variance:

$$\mathrm{Var}(R(y_j)) = \underbrace{\mathrm{Var}(\mathbb{E}[R(y_j) \mid \mathcal{I}_{\mathrm{CoT}}(c_j)])}_{\text{Variance of conditional expectation}} + \underbrace{\mathbb{E}[\mathrm{Var}(R(y_j) \mid \mathcal{I}_{\mathrm{CoT}}(c_j))]}_{\text{Expectation of conditional variance}}.$$

First term:

$$\mathbb{E}[R(y_j) \mid \mathcal{I}_{\mathrm{CoT}}(c_j)] = \begin{cases} \alpha & \text{if } \mathcal{I}_{\mathrm{CoT}}(c_j) = 1 \\ \beta & \text{if } \mathcal{I}_{\mathrm{CoT}}(c_j) = 0 \end{cases}.$$

The random variable $\mathbb{E}[R(y_j) \mid \mathcal{I}_{\mathrm{CoT}}(c_j)]$ has variance:

$$\mathrm{Var}(\mathbb{E}[R(y_j) \mid \mathcal{I}_{\mathrm{CoT}}(c_j)]) = (\alpha-\beta)^2 p_c(1-p_c).$$

Second term:

$$\mathrm{Var}(R(y_j) \mid \mathcal{I}_{\mathrm{CoT}}(c_j)) = \begin{cases} \alpha(1-\alpha) & \text{if } \mathcal{I}_{\mathrm{CoT}}(c_j) = 1 \\ \beta(1-\beta) & \text{if } \mathcal{I}_{\mathrm{CoT}}(c_j) = 0 \end{cases},$$

so its expectation is:

$$\mathbb{E}[\mathrm{Var}(R(y_j) \mid \mathcal{I}_{\mathrm{CoT}}(c_j))] = p_c\alpha(1-\alpha) + (1-p_c)\beta(1-\beta).$$

Thus:

$$\sigma^2 = (\alpha-\beta)^2 p_c(1-p_c) + p_c\alpha(1-\alpha) + (1-p_c)\beta(1-\beta). \tag{14}$$

Substituting μ and σ into equation 12 and equation 13, we have

$$\mathbb{E}[\hat{A}(y_i) \mid \text{correct CoT}] \to \frac{(1-p_c)(\alpha-\beta)}{\sqrt{(\alpha-\beta)^2 p_c(1-p_c) + p_c\alpha(1-\alpha) + (1-p_c)\beta(1-\beta)}},$$

$$\mathbb{E}[\hat{A}(y_i) \mid \text{incorrect CoT}] \to \frac{-p_c(\alpha-\beta)}{\sqrt{(\alpha-\beta)^2 p_c(1-p_c) + p_c\alpha(1-\alpha) + (1-p_c)\beta(1-\beta)}}.$$

An ideal pre-training on a high-capacity model could help to ensure that $\alpha \to 1$ and $\beta \to 0$ at the beginning of RLVR. In this condition, we have the following advantage estimates:

$$\mathbb{E}[\hat{A}(y_i) \mid \text{correct CoT}] \to \sqrt{\frac{1-p_c}{p_c}}, \quad \mathbb{E}[\hat{A}(y_i) \mid \text{incorrect CoT}] \to -\sqrt{\frac{p_c}{1-p_c}}.$$

In this ideal scenario, the role of human would be to prepare a comprehensive and diverse set of questions and answers, leveraging RLVR to automatically incentivize the model's reasoning capabilities. However, in practice—the "unideal case"—it is often necessary to first fine-tune the base LLM to align its output with a proper reasoning distribution before applying RLVR.

Discussions on Key Observations in RLVR Grounded in our theoretical analysis, we can now provide our unique explanations for several previously elusive yet important observations reported in DeepSeek-R1 (Guo et al., 2025).

Our Explanation of the Observation "*DeepSeek-R1-Zero achieved remarkable Pass@K performance on AIME 2024 but encountered challenges such as poor readability and language mixing.*": Even DeepSeek-V3 (Liu et al., 2024) cannot guarantee ideal conditions where $\alpha \to 1, \beta \to 0$. As a result, cold-start data is required to rectify prior logic biases, motivating the R1 approach.

Our Explanation of the Observation "*The R1-Zero approach did not work well for the 32B dense model, yet distillation can be very effective.*": Key factors such as (p_c, α, β) for the 32B base model are in an even worse state, causing pure RLVR to converge to suboptimal local solutions. Based on our analysis, the key to effective reasoning lies in learning correct CoTs. Therefore, the distillation approach can efficiently teach an LLM how to reason properly.

A.6 Additional Details in Reproducing DAPO Training

Our reproduction was conducted on 32 AMD MI300X GPUs using the VERL framework (Sheng et al., 2025), and ran for over two weeks. Our run did not fully reproduce the $Pass@1$ accuracy above 50% as reported by Yu et al. (2025), while we reached a comparable performance of around 44% $Pass@1$, in line with a third-party reproduction (Chen et al., 2025a). We use the same verifier introduced in Section 3 to assess the correctness of both training and evaluation rollouts.

In addition to the performance evolution on fully optimized training questions highlighted in Figure 4 of the main paper, we include performance evolution on hard questions and more continuous validation performance in Figure 10 to provide more comprehensive information. The additional observations are consistent with what we have introduced in the main paper. It is natural to observe that RLVR begins to incentivize correct reasoning from the very beginning, as evidenced by increased $P(CC|CA)^{(q)}$ values in the early training steps shown in Figures 10(a) and 10(b). These incentivized reasoning capabilities translate into improved generalization on unseen questions, as demonstrated by notable gains in CoT-Pass@K on AIME 2024 within the first 20 training steps in Figure 10(c). Note that each training step here corresponds to one round of PPO-style optimization (Schulman et al., 2017), which includes 16 gradient updates, according to the DAPO training script. Thus, we see that correct reasoning abilities begin to generalize after only a few gradient updates.

Furthermore, the incentivization of correct reasoning on training questions appears to be a continuous process, as reflected by the steady increase in the mean of $P(CC|CA)^{(q)}$ throughout training, for both easy and hard questions. Meanwhile, we again observe that $P(CA)^{(q)}$ (equivalent to $Pass@1^{(q)}$) is an unreliable metric, particularly for easy training questions. As shown in Figure 10(a), the distribution of $P(CA)^{(q)}$ becomes highly skewed toward 1.0 after 180 steps, misleadingly suggesting that most questions are perfectly solved. However, examining the distribution of $P(CC|CA)^{(q)}$ reveals that a substantial fraction of responses still contain flawed reasoning. We suspect this is one of the reasons behind the difficulty of achieving strong results with Qwen2.5-32B using the R1-zero approach.

For both easy and hard training questions, improving $P(CC|CA)^{(q)}$ seems to be a slow and challenging process. Since our analysis shows that enhancing correct CoTs is key to improving reasoning capabilities, we believe that future research should explore novel mechanisms to accelerate the improvement of $P(CC|CA)^{(q)}$, thereby enhancing both the efficiency and effectiveness of RLVR.

A.7 Discussions

Call for Live, Challenging Benchmarks Static benchmarks developed prior to the release of modern base models are increasingly susceptible to contamination risks, potentially undermining the reliability of observed improvements. In response, we emphasize the need for *live benchmarks* that evolve over time, as suggested in recent studies (Jain et al., 2025; White et al., 2025). Additionally, we agree with the viewpoint of Yao (2025) that future research advancements may rely more on designing new evaluations, benchmarks, and environments.